

In the ZTE ZXA10 C600, the SNMP **ifIndex** for physical ports, including GPON ports, is classified as **Type 1** 1, 2. The value is a 32-bit integer that encodes the hardware hierarchy—Rack, Shelf, Slot, and Port—into specific bit positions 2.

Bit Breakdown for Type 1 (Physical Interfaces)

The ifIndex is structured from the most significant bit (MSB) to the least significant bit (LSB) as follows 2:

Bits,Field,Description / Typical Value

31–28,Type,Fixed at 1 for physical interfaces.

27–24,Rack ID,Typically fixed at 1.

23–16,Shelf ID,Typically fixed at 1.

15–8,Slot ID,The numerical ID of the line card slot.

7–0,Port ID,The numerical ID of the GPON or Ethernet port.

Bit-Shift Formula

To calculate the decimal or hexadecimal ifIndex value, use the following bit-shift and OR operations 2:

ifIndex = (1 << 28) | (RackID << 24) | (ShelfID << 16) | (SlotID << 8) | PortID

(Note: In most ZXA10 C600 configurations, RackID and ShelfID are both 1) 2.

Concrete Example: Slot 1, Port 1

To find the ifIndex for the first port in the first slot:

1. **Type (1):** $1 \ll 28 = 0x10000000$
2. **Rack (1):** $1 \ll 24 = 0x01000000$
3. **Shelf (1):** $1 \ll 16 = 0x00010000$
4. **Slot (1):** $1 \ll 8 = 0x00000100$
5. **Port (1):** $1 = 0x00000001$

Calculation: $0x10000000 | 0x01000000 | 0x00010000 | 0x00000100 | 0x00000001 = 0x11010101$

- **Hexadecimal:** 0x11010101
- **Decimal:** 285278465

As a comparison, the index for **PON port 2 in Slot 1** would be 0x11010102 3.

Would you like to see how logical sub-indices (such as Vport or SPORT IDs) are calculated for these GPON ports?